

PHASE TRANSITIONS IN A COMPLEX NETWORK

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ABSTRACT. We study a mean field model of a complex network, focusing on edge and triangle densities. Our first result is the derivation of a variational characterization of the entropy density, compatible with the infinite node limit. We then determine the optimizing graphs for small triangle density and a range of edge density, though we can only prove they are local, not global, maxima of the entropy density. With this assumption we then prove that the resulting entropy density must lose its analyticity in various regimes. In particular this implies the existence of a phase transition between distinct heterogeneous multipartite phases at low triangle density, and a phase transition between these phases and the disordered phase at high triangle density.

1. INTRODUCTION

Exponential random graph models are a well known class of complex networks; see [Ne] and references therein. Using the language of statistical mechanics they are mean field models, in the grand canonical ensemble, with a variety of possible many-body interactions appropriate to the model's use. Phase transitions, which require an infinite node limit, have been proven for them [CD, RY] using the recently developed ‘graphon’ formalism [Lov] in place of the infinite volume limit formalism [R1, R2] used in statistical mechanics. Exponential random graph models are mean field models and therefore the analogues of the various statistical mechanics ensembles (microcanonical, grand canonical, pressure, ...) which are equivalent in the infinite volume limit for particle systems with short range interactions [R1], need not be equivalent in these mean field models; see for instance [TET]. (Equivalence of ensembles is discussed further in the Conclusion.) In this work we use the microcanonical ensemble of one of the best known exponential random graph models, one originally formulated by Strauss [St], and give evidence of phase transitions which are not as accessible in the grand canonical ensemble. The transitions previously analyzed for a wide class of exponential random graphs are similar to liquid/gas transitions in that they are transitions between graphs of similar character, of the same (fluid-like) phase [RY], while the transitions we focus on in the microcanonical ensemble are analogous to solid/solid transitions, transitions between graphs of different phases. (See [AR] for a more primitive grand canonical analysis of these phases.)

We need some network notation. Consider the set $\hat{G}^n$ of simple graphs G with set $V(G)$ of (labelled) vertices, edge set $E(G)$ and triangle set $T(G)$, where the cardinality $|V(G)| = n$. (‘Simple’ means the edges are undirected and there are no multiple edges or loops.) Think of an unordered pair of vertices as a point in an abstract space, an edge as a particle that may occupy that point, and a triangle as a many-body interaction energy associated with its

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edges, so the microcanonical partition function, $Z_{e,t}^{n,\delta}$, is the number of simple graphs such that:

$$(1) \quad e(G) \equiv \frac{|E(G)|}{\binom{n}{2}} \in (e - \delta, e + \delta) \quad \text{and} \quad t(G) \equiv \frac{|T(G)|}{\binom{n}{3}} \in (t - \delta, t + \delta).$$

Graphs in $\cup_{n \geq 1} \hat{G}^n$ are known to have edge and triangle densities, (e, t) , dense in the region R in the e, t -plane bounded by three curves, $c_1 : (e, e^{3/2})$, $0 \leq e \leq 1$, the line $l_1 : (e, 0)$, $0 \leq e \leq 1/2$ and a certain scalloped curve $(e, h(e))$, $1/2 \leq e \leq 1$, lying above the curve $(e, e(2e - 1))$, $1/2 \leq e \leq 1$, and meeting it when $e = e_k = k/(k + 1)$, $k \geq 1$; see [PR] and references therein, and Figure 1.

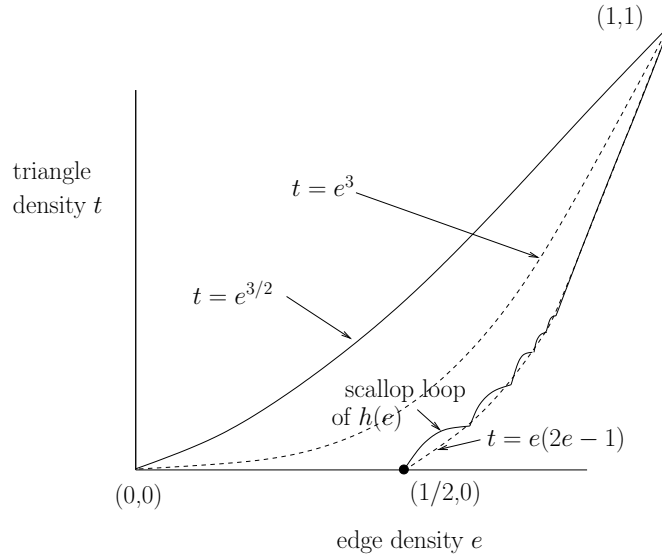


FIGURE 1. The microcanonical phase space R , outlined in solid lines

We are interested in the relative density of graphs in R , more precisely in the entropy, the exponential rate of growth of the number of graphs as n grows, as follows. First consider

$$(2) \quad s_{e,t}^{n,\delta} = \frac{\ln(Z_{e,t}^{n,\delta})}{n^2}, \text{ and then } s_{e,t} = \lim_{\delta \rightarrow 0^+} \lim_{n \rightarrow \infty} s_{e,t}^{n,\delta}.$$

(The existence of the double limit will be proven.) We will measure the growth rate by the entropy density $s_{e,t}$, and the main question of interest for us is the existence of phase transitions (*i.e.* lack of analyticity of $s_{e,t}$) near the lower boundary of R in Figure 1. The lower boundary consists of the scalloped curve together with the ‘first scallop’, the line from $(0, 0)$ to $(1/2, 0)$.

We now need to review some notation and results concerning graphons, as recently developed in [LS1, LS2, BCLSV, BCL, LS3]. See also the recent book [Lov].

2. GRAPHONS

Consider the set $\mathcal{W}$ of all symmetric, measurable functions

$$(3) \quad g : (x, y) \in [0, 1]^2 \rightarrow g(x, y) \in [0, 1].$$

Think of each axis as a continuous set of vertices of a graph. For a graph $G \in \hat{G}^n$ one associates

$$(4) \quad g^G(x, y) = \begin{cases} 1 & \text{if } ([nx], [ny]) \text{ is an edge of } G \\ 0 & \text{otherwise,} \end{cases}$$

where $[y]$ denotes the smallest integer greater than or equal to y . For $g \in \mathcal{W}$ and simple graph H we define

$$(5) \quad t(H, g) \equiv \int_{[0,1]^\ell} \prod_{(i,j) \in E(H)} g(x_i, x_j) dx_1 \cdots dx_\ell,$$

where $\ell = |V(H)|$, and note that for a graph G , $t(H, g^G)$ is the density of graph homomorphisms $H \rightarrow G$:

$$(6) \quad \frac{|\text{hom}(H, G)|}{|V(G)|^{|V(H)|}}.$$

We define an equivalence relation on $\mathcal{W}$ as follows: $f \sim g$ if and only if $t(H, f) = t(H, g)$ for every simple graph H . Elements of $\mathcal{W}$ are called “graphons”, elements of the quotient space $\tilde{\mathcal{W}}$ are called “reduced graphons”, and the class containing $g \in \mathcal{W}$ is denoted $\tilde{g}$. Equivalent functions in $\mathcal{W}$ differ by a change of variables in the following sense. Let Σ be the space of measure preserving maps $\sigma : [0, 1] \rightarrow [0, 1]$, and for f in $\mathcal{W}$ and $\sigma \in \Sigma$, let $f_\sigma(x, y) \equiv f(\sigma(x), \sigma(y))$. Then $f \sim g$ if and only if there exist σ, σ' in Σ such that $f_\sigma = g_{\sigma'}$ almost everywhere; see Cor. 2.2 in [BCL]. The space $\mathcal{W}$ is compact with respect to the ‘cut metric’ defined as follows. First, on $\mathcal{W}$ define:

$$d_\bullet(f, g) \equiv \sup_{S, T \subseteq [0,1]} \left| \int_{S \times T} [f(x, y) - g(x, y)] dx dy \right|.$$

Then on $\tilde{\mathcal{W}}$ define the cut metric by:

$$\tilde{d}_\bullet(\tilde{f}, \tilde{g}) \equiv \inf_{\sigma, \sigma' \in \Sigma} d_\bullet(f_\sigma, g_{\sigma'}).$$

We will use the fact, which follows easily from Lemma 4.1 in [LS1], that the cut metric is equivalent to the metric

$$(7) \quad \delta_{\text{hom}}(\tilde{f}, \tilde{g}) \equiv \sum_{j \geq 1} \frac{1}{2^j} |t(H_j, f) - t(H_j, g)|,$$

where $\{H_j\}$ is a countable set of simple graphs, one from each graph-equivalence class. Also note that if each vertex of a finite graph is split into the same number of ‘twins’, each connected to the same vertices, the result stays in the same equivalence class, so for a convergent sequence $\tilde{g}^{G_j}$ one may assume $|V(G_j)| \rightarrow \infty$.

The value of this graphon formalism here is that one can use large deviations on graphs with independent edges, proven in [CV], to give an optimization formula for $s_{e,t}$, which allows us to analyze $s_{e,t}$ near the graphons of minimal triangle density, the lower boundary of R in Figure 1. We next use the large deviations rate function for graphs with independent edges to give a variational characterization for the entropy density. (There is a variational characterization in [CD] of the free energy density in the grand canonical ensemble.)